Gold and Silver

Economic Drivers and Price Prediction Across Crises and Elections

Sanskriti & Anisha

Research Question

How do macroeconomic and market variables influence gold and silver price movements, and does economic policy uncertainty affect their ability to predict price trends?

Tractable Data & Data Retrieval

Data sources:

- World Bank
- Federal Reserve Economic Data
- Kaggle – daily gold and silver prices

Dataset from these sources are standardized and merged by dates (daily) omitting weekends to focus solely during trading periods.

Why Tractable:
Structured time series datasets, easy merging across dates, interpretable economic variables

EDA / Plan for project

Macroeconomic indicators:

1. Inflation data (CPI):
   – Captures changes in purchasing power, making gold and silver attractive as inflation hedges.
2. Interest rates:
   – Higher rates tend to reduce their appeal, while lower rates can increase demand.
3. US dollar index:
   – Reflects the strength of the US dollar, in which gold and silver are priced globally.

Economic Policy Uncertainty Index

Economic Policy Uncertainty Index (daily):

- Frequency of newspaper articles mentioning:
  - –economy
  - –policy
  - –uncertainty

By integrating uncertainty into a predictive modeling framework, the analysis aims to capture how changing political and economic conditions reshape market behavior.

ML Methods

Linear regression

Provides a baseline to estimate direct, interpretable relationships between macro variables and prices.

Random Forest

Captures non-linear relationships between macroeconomic and political variables. Provides feature importance, helping identify key drivers under different conditions.

XGBoost

Builds a strong predictive model by combining many smaller models and correcting errors step by step.

Implications for Stakeholders

1. Portfolio Managers and Investors: Used as a hedging risk, decide how to allocate between stocks, commodities, alternative assets

2. Retail investors & Households: marketed as a safe investment, people invest during inflation fears or crisis

3. Central bank & Government: Holds gold reserves, monitors inflation expectations, currency strength, global financial stability

4. Mining Industry: Revenue depends on gold silver prices, countries depend on commodity exports

5. Traders & Financial Analysts: Trade short term movements, build predictive models

Ethical, legal, societal implications

1. Limitations of Financial Prediction: Financial markets are uncertain and influenced by unpredictable events, over reliance on predictions can lead to risky financial decisions

2. Human rights abuse: In some regions, mining industries are linked to labor exploitation, unsafe working conditions and environmental harm

3. Correlation vs causation: Observed relationships do not imply direct causality, Important to find patterns

4. Unequal access to financial knowledge: Advanced data and modelling tools are not equally accessible, institutional investors have more resources than individuals

5. Market Narratives: "Gold is a safe haven" is simplified narrative, Misleading assumptions can affect financial decision